

The Impact of Ability Grouping on Academic Performance: An Empirical Analysis

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I. Introduction

Many educational systems across the world have utilized ability grouping across subjects. The enhancement or hindering of learning outcomes due to ability grouping has been long debated across educational systems. Proponents of ability grouping argue that ability grouping helps learning outcomes through individualized instruction to students depending on the student's skill level and knowledge of the subject. Critics argue that it provides inequality of education by concentrating resources to groups of higher performing students, essentially "leaving behind" students who have performed poorly in the past. This tension between the two sides has led ability grouping to be heavily debated in educational policy.

Prior research has found conflicting results on the subject. Studies from Gamoran (Is Ability Grouping Equitable, 1992) have shown that ability grouping raises performance for high performing students while widening the achievement gap for lower performing students. A study from OECD (Education at a Glance, 2019) showed that the effect is heavily dependent on whether grouping occurs within classrooms, across subjects, or at school level. Additionally, the degree of resource equality made a difference in the results. Results of most studies on the subject show that school-level characteristics play a major role in this, such as funding and teacher quality.

My goal with this paper is to contribute to the discussion by estimating the effect of ability grouping on student performance in math, reading, and science using data from the PISA (Programme for International Student Assessment) in 2009. The PISA was an assessment given to schools involving 65 countries focusing on outcomes of math, reading, and science scores. The analysis is comparing schools that use ability grouping in all subjects and in some subjects to schools that do not use any ability grouping. Using ability grouping among subjects or not is a choice, so to address selection bias the regression model includes a plethora of both school-level and student-level controls. The school-level controls are designed to cover sources of funding, school size, quality of teachers, disciplinary climate, and educational resources. Student-level controls are designed to cover student demographics, attendance, parental education, and socioeconomic status. Additionally, a country by grade fixed effect is included to keep the comparison among schools within the same country and students within the same grade.

My findings suggest that ability grouping does lead to a slight increase in PV1 scores, therefore an increase in academic performance, but not by as much as expected. While a naive comparison would suggest a larger effect, controlling for school-level, student-level, country-level, and grade-level effects shows a minor effect of ability grouping. Instead, PV1 assessment

scores are impacted by many underlying differences between schools and students. The results suggest that broader socioeconomic and educational conditions play a more decisive role in student performance.

II. Data

The PISA included many variables that proved useful in my regression model. The most important variables that were already present inside of the PISA were PV1 scores for math, reading, and science. I used these as outcome variables inside three separate models. Another important variable listed in the PISA data was ABGROUP, which is an ability grouping measure where schools listed 1 if they had ability grouping for all subjects, 2 if they had ability grouping in some subjects, and 3 if they had ability grouping for no subjects. I used this variable to create three dummy variables representing ability grouping for all subjects (all_abgroup), ability grouping for some subjects (some_abgroup), and ability grouping for no subjects (no_abgroup). The dummy variables all_abgroup and some_abgroup were included in all three regression models as the treatment to be compared to schools that had no ability grouping.

After the treatment and control were established, control variables were made on the school level to address the inherent selection bias of ability grouping. Schools that have ability grouping of subjects may have access to better resources, funding, and student compositions, so it is important to control for these potential advantages to better understand the true impact of ability grouping on test scores. Firstly, the differences in funding were addressed using variables which were already present within the PISA, which include the percentage of government funding, student-fee funding, and benefactor funding which schools receive. Secondly, another variable present within the PISA was SCHSIZE, which accounts for total enrollment of students within schools and should control for differences in school size. Two more school controls already present within the PISA were used, DISCLIMA and SCMATEDU, which is the disciplinary climate within schools and quality of educational resources. Finally, dummy variables were created for private vs public schools and shortages of qualified teachers.

After creating school-level controls, the differences among students were addressed since students may have different test scores for many different reasons. The only variable used directly from the PISA includes ESCS, which is a measure of a student's economic, social, and cultural status. This variable included errors of outliers and incorrect entries which had to be removed, as was the problem with many variables. All other controls for the student-level were dummy variables created from the dataset. The first control considered for this was immigration status, as immigrants are at a disadvantage academically. Dummy variables were created for first-generation and second-generation immigrants. Secondly, attendance was considered since missing more classes would likely lead to much worse test scores. Dummy variables were created for students who had no absences, very little absences, more absences, and a lot of absences. Students who have access to better resources outside of school likely have better test scores, so a dummy variable (studyplace) was created to address this. The variable studyplace

addresses whether a student has access to a proper study place outside of school, which can help control for differences in house size and potentially overcrowding at home.

Another student-level variable created was outofschoollessons, which is 1 if a student receives out of school lessons for any subject. Next a variable was created for females since there are inherent differences between males and females. Finally, dummy variables were created to address the amount of education a student's parent has since parents with more education are more likely to push better education onto their child. Additionally, a more educated parent likely makes more money, so these variables control for more differences in student home life. The variables included parents that finished less than 12 years of education (parent_somehs), parents that finished 12 years of education (parent_hs), parents that finished more than 12 years of education but less than 16 (parent_somecollege), parents that finished 16 years of education (parent_collegedegree), and parents who finished 18 years of schools (parent_graduatedegree).

III. Empirical Approach

The samples of students and schools include a meaningful variation. The sample is not perfectly balanced, as schools with ability grouping tend to be larger and have better resources, with higher ESCS scores among students. Meanwhile, ungrouped schools display lower funding and a worse disciplinary climate. Although they differ, you can observe in Table 1 that the control and treatment groups are comparable.

Table 1 Estimates the mean of some of the Treatment group (Any ability grouping of subjects) and Control group's (No ability grouping of subjects) characteristics to estimate comparability before calculations.

	Variable	Treatment Group Mean	Control Group Mean	Difference (T-C)
Student Level	Female (%)	37.92%	38.72%	(0.8%)
	Age	15.78	15.79	(.01)
	Grade Level	9.62	9.54	0.08
	Foreign Language At Home (%)	12.27%	15.07%	(2.80)
	Economic, Social, and Cultural Status	-0.28	-0.40	0.12
School level	Total Enrollment	829.34	960.84	(131.50)
	Student-Teacher Ratio	15.71	16.91	(1.20)
	Private School (%)	18.44%	19.71%	(1.27%)

Disciplinary Climate	0.76	0.74	0.02
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Table 1 shows that the two groups are highly comparable in both student composition and school qualities before the regression takes place. This supports the validity of the regression models and will reduce the likelihood of treatment effects being driven by differences among the groups.

To start my regression models, I wanted to estimate the naive comparison between students in schools with ability grouping and students in schools without. I made the three naive regression models: ($PVISCIE = \beta_0 + \beta_{1all_abgroup} + \beta_{2some_abgroup} + \epsilon$), ($PVIREAD = \beta_0 + \beta_{1all_abgroup} + \beta_{2some_abgroup} + \epsilon$), and ($PVIMATH = \beta_0 + \beta_{1all_abgroup} + \beta_{2some_abgroup} + \epsilon$). In the models, PVIMATH, PVIREAD, and PVISCIE are the PISA plausible values. All_abgroup and some_abgroup are dummy variables for ability grouping in all subjects and some subjects, with no ability grouping as the omitted reference. It is expected that these estimates are to be highly biased, since ability grouping among subjects is a choice heavily biased on different circumstances and characteristics.

In my second set of models, I added school-level controls to see the impact on the coefficients of ability grouping. The three regression models are as follows: ($PVISCIEs = \beta_0 + \beta_{1all_abgroup} + \beta_{2some_abgroup} + Ws'\gamma + \epsilon_s$), ($PVIREADs = \beta_0 + \beta_{1all_abgroup} + \beta_{2some_abgroup} + Ws'\gamma + \epsilon_s$), and ($PVIMATHs = \beta_0 + \beta_{1all_abgroup} + \beta_{2some_abgroup} + Ws'\gamma + \epsilon_s$). In the models, s indexes schools, while PVIMATH, PVIREAD, and PVISCIE are the PISA plausible values. All_abgroup and some_abgroup are dummy variables for ability grouping in all subjects and some subjects, with no ability grouping as the omitted reference. Finally, Ws is a vector of school-level controls: school's total enrollment, public/private, school funding, qualified-staff shortage index, disciplinary climate, and quality of school's educational resources.

These models should be more accurate than the naive comparison but are likely to still be heavily biased. This is because student characteristics have a major impact on PV1 assessment scores, so my final three regression models will include student-level controls, as well as a country by grade fixed effect to narrow the comparison to students within the same country and within the same grade. This fixed effect is very important because the PISA assessments included 65 countries, and events that happen within a country at the time could heavily influence assessment scores. Additionally, a fixed effect for grades will help hold fixed many variables about the students, including past educational experience and changes in educational standards.

My final three regression models were created to evaluate the impact of test scores of three different subjects when controlling for student-level, school-level, and country-by-grade-level bias. The model are as presented: ($PVISCIEisc = \beta_0 + \beta_{1all_abgroup} + \beta_{2some_abgroup}$

+ $Xi'\delta + Ws'\gamma + \alpha_{cg} + \varepsilon_{isc}$), ($PV1READisc = \beta_0 + \beta_1all_abgroup + \beta_2some_abgroup + Xi'\delta + Ws'\gamma + \alpha_{cg} + \varepsilon_{isc}$), and ($PV1MATHisc = \beta_0 + \beta_1all_abgroup + \beta_2some_abgroup + Xi'\delta + Ws'\gamma + \alpha_{cg} + \varepsilon_{isc}$). In the models, i indexes students, s indexes schools, c indexes countries, and g indexes grades. PV1MATH, PV1READ, and PV1SCIE are the PISA plausible values.

All_abgroup and some_abgroup are dummy variables for ability grouping in all subjects and some subjects, with no ability grouping as the omitted reference. Xi is a vector of student-level controls: gender, ESCS, parental education, immigrant status, home possesses study place, student absenteeism, language spoken at home, and out of school lessons. Ws is a vector of school-level controls: school's total enrollment, public/private, school funding, qualified-staff shortage index, disciplinary climate, and quality of school's educational resources. α_{cg} represents country-by-grade fixed effects. Additionally, standard errors are clustered at the school level.

IV. Results

Table 2 Estimates the naive comparison of ability grouping in all subjects and some subjects to no subjects without controlling for any bias. The data presented includes coefficients as well as Test-statistics in parenthesis. Asterisks represent statistical significance at the 90th percentile (*), 95th percentile (**), and 99th percentile (***). The coefficients are interpreted in units of PV1 score.

	PV1 Math	PV1 Read	PV1 Scie
Ability grouping in all subjects	18.98*** (43.47)	19.13*** (44.30)	18.75*** (43.04)
Ability grouping in some subjects	34.01*** (82.31)	30.18*** (73.85)	31.93*** (77.46)
School-level Controls	N	N	N
Student Level Controls	N	N	N
Country by Grade fixed effect	N	N	N

Table 2 shows that without controlling for any bias, students in schools that have ability grouping in all subjects scored on average over 18 points higher on the PV1 assessments in math, reading, and science compared to students in schools with no ability grouping. Students in schools that have ability grouping in some subjects did even better, with over 30 points in every subject higher on the PV1 assessments. We know that this is heavily biased, so school-level controls will be added to control for bias on the school level.

Table 3 Estimates the average effect of ability grouping in all subjects and some subjects controlling for school-level bias. The data presented includes coefficients as well as Test-statistics in parenthesis. Asterisks represent statistical significance at the 90th percentile (*), 95th percentile (*), and 99th percentile (**). The coefficients are interpreted in units of PV1 score.

	PV1 Math	PV1 Read	PV1 Scie
Ability grouping in all subjects	16.37*** (34.03)	17.79*** (37.35)	16.34*** (33.93)
Ability grouping in some subjects	29.51*** (65.46)	27.33*** (61.25)	28.26*** (62.63)
Total Enrollment	.0133*** (57.69)	.0123*** (54.03)	.0114*** (49.73)
Student Fee Funding	.101*** (5.89)	.0935*** (5.51)	.0292 (1.70)
Private	28.51*** (58.83)	22.77*** (47.47)	26.82*** (55.30)
More Shortage of qualified teachers	.005 (0.01)	-3.41*** (-7.61)	-2.30*** (-5.07)
Disciplinary Climate	9.15*** (54.96)	10.89*** (66.06)	8.72*** (52.34)
Quality of Schools Educational Resources	23.64*** (144.09)	20.30*** (125.03)	22.55*** (137.33)
School-level Controls	Y	Y	Y
Student Level Controls	N	N	N
Country by Grade fixed effect	N	N	N

What stands out in this table is that when adding controls for school-level bias, you can see that the coefficients for both ability groupings were being overstated in the naive comparison. Variables which stand out include disciplinary climate and quality of school's educational resources because both having a better disciplinary climate and having better quality of

educational resources can greatly boost PV1 assessment scores. Additionally, a shortage of qualified teachers tends to decrease PV1 assessment scores, excluding PV1 math scores because it was no different from 0. It can also be observed that when controlling for a school's private or public status, private schools tend to do much better on the PV1 assessments. This is likely due to characteristics within students who attend private school since they likely have advantages to students who do not.

This is why it is important to add student-level characteristics to further reduce bias within the model. Additionally, the comparison in Table 4 is between students in schools that have ability grouping in subjects and students in schools that do not across many countries. To narrow the comparison, the country by grade fixed effect is necessary.

Table 5 Estimates the average effect of ability grouping in all subjects and some subjects compared to ability grouping in no subjects on PV1 assessment scores with school-level controls, student-level controls, and a country by grade fixed effect. The data presented includes coefficients as well as Test-statistics in parenthesis. Asterisks represent statistical significance at the 90th percentile (*), 95th percentile (**), and 99th percentile (***). The coefficients are interpreted in units of PV1 score.

	PV1 Math	PV1 Read	PV1 Scie
Ability grouping in all subjects	4.123** (2.74)	2.614 (1.86)	3.146* (2.22)
Ability grouping in some subjects	4.256** (2.97)	3.731** (2.59)	3.590* (2.52)
Total Enrollment	0.00881*** (9.16)	0.0105*** (11.75)	0.00894 (10.53)
Student Fee Funding	0.178** (3.08)	0.110* (2.06)	0.121* (2.20)
Private	-0.699 (-0.40)	2.453 (1.53)	-0.867 (-0.51)
More Shortage of qualified teachers	-4.255** (-3.17)	-5.082*** (-3.74)	-4.457** (-3.21)
Disciplinary Climate	7.870*** (31.95)	10.82*** (43.98)	8.521*** (35.68)
Quality of Schools Educational Resources	2.919*** (5.44)	2.549*** (4.87)	2.387*** (4.47)
<i>First-generation immigrant</i>	4.780* (2.51)	7.471*** (3.68)	3.654 (1.80)

<i>Second-generation immigrant</i>	-0.0164 (-0.01)	3.000* (2.23)	-3.035* (-2.13)
<i>Student absent a lot</i>	-29.11*** (-13.71)	-28.68*** (-13.14)	-27.85*** (-12.97)
<i>Student access to a study place</i>	6.258*** (6.98)	6.155*** (6.52)	6.784*** (8.02)
<i>Student out of school lessons</i>	-15.48*** (-30.27)	-13.74*** (-26.56)	-15.47*** (-29.71)
<i>Student economic, social, and cultural status</i>	24.92*** (73.31)	23.39*** (72.07)	23.85*** (75.20)
<i>Female</i>	-1.673 (-1.79)	4.159*** (4.48)	-0.688 (-0.75)
<i>Constant</i>	332.5*** (11.31)	358.5*** (12.17)	396.7*** (14.60)
School-level Controls	Y	Y	Y
Student Level Controls	Y	Y	Y
Country by Grade fixed effect	Y	Y	Y
Sample Size	231,636	231,636	231,636

Table 5 has a much different outcome from the naive comparison made in table 2. Ability grouping of all subjects is shown to only have an increase of 4.123 points in PV1 math scores, a statistically insignificant 2.614 points in PV1 reading scores, and 3.146 points in PV1 science scores, while ability grouping of some subjects has slightly higher increases of 4.256 points in math, 3.731 points in reading, and 3.590 points in science. These results show that the coefficient for all subjects grouped and some subjects grouped were being heavily overstated by student-level, school-level, country-level, and grade-level characteristics.

At the school level, it can be observed that a one unit increase in student-fee funding can increase PV1 scores by as much as 0.178 points. Additionally, the coefficients for private schools have become insignificant across all three subjects, suggesting that the previous impact in table 4 of private schooling was largely due to omitted variables of students. Students in schools with

more shortages of qualified teachers had over 4.2 points lower PV1 scores across subjects, with as high as 5.082 points less in the reading PV1 assessment. A one unit increase in the disciplinary climate index also had a large increase in scores across subjects, suggesting that a better disciplinary climate in schools will lead to better performance of students. Finally, an increase in the quality of a school's educational resources adds over a 2-point increase across subjects.

At the student level, the results were very interesting and revealed that a student's PV1 score is heavily impacted by variables at the student level. Results show students who were absent a lot did much worse on their PV1 exams across subjects, with a 29.11 decrease in math, 28.68 decrease in reading, and 27.85 decrease in science, all of which were statistically significant at the 99th percentile. Additionally, students who have access to a study place at home had over 6 points higher across subjects than students who did not, suggesting that an overcrowded, potentially disruptive household has a negative effect on a student's performance. Initially students who receive out of school lessons was added to control for students who received more education potentially yielding higher test scores, but the results show much worse performance on the PV1 assessments across subjects, majorly due to students who are required to take tutoring due to previous poor performance.

Regarding gender, being female had insignificant results in math and science. However, females had 4.159 higher PV1 scores on average in reading compared to males. An interesting result regarding immigration is that in math, first generation immigrants had 4.780 points higher than non-immigrants and in reading they had 7.471 points higher than non-immigrants. Additionally, second-generation immigrants on average scored 3 points higher in reading and 3 points lower in science. You would potentially expect immigrants to do worse on assessments due to language barriers, but this is consistent with the immigrant paradox, which states that recent immigrants typically do better academically than more established immigrants and non-immigrants. This is likely because first-generation immigrants usually enter a country in search of opportunity and bring with them a strong work ethic and discipline.

Finally, and potentially the most important school level control, is a student's economic, social, and cultural status (ESCS). A one unit increase to a student's ESCS had substantial increases to PV1 assessment scores, with a 24.92-point increase in math, a 23.39-point increase in reading, and a 23.85-point increase in science, all of which are statistically significant at the 99th percentile.

V. Conclusion

The findings suggest that ability grouping did in fact have a positive increase in PV1 scores across subjects, but it was not a substantial increase. PV1 assessment scores are more heavily affected by a school's funding, quality of educational resources, and disciplinary climate, and more importantly characteristics of a student. Students with a better economic, social, and cultural background tend to have much better assessment scores. Additionally, attendance was found to be vital to a student's success on assessments. With that being said, ability grouping can

be beneficial to an educational system, but there are other, more significant ways that schools can increase a student's success.

This paper supports both the argument for and against ability grouping. On one hand, there was a slight increase in test scores across students in schools with ability grouping as opposed to students without ability grouping. However, it often boils down to other characteristics such as the economic background of students within school's which ability group and advantages in funding, qualified staff, and educational resources which schools that do not have ability grouping often lack. Unlike many of the studies published on the topic, this paper includes schools which had both ability grouping in all subjects and ability grouping in some subjects, so its impact can be observed on a spectrum. Additionally, the PISA used data from many countries and many schools across these countries, so the use of a country-by-grade fixed effect can isolate the effect of ability grouping from events which happen in each country. The findings align most with recent research that suggests that the impact of ability grouping is miniscule when accounting for school-level and student-level characteristics.

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